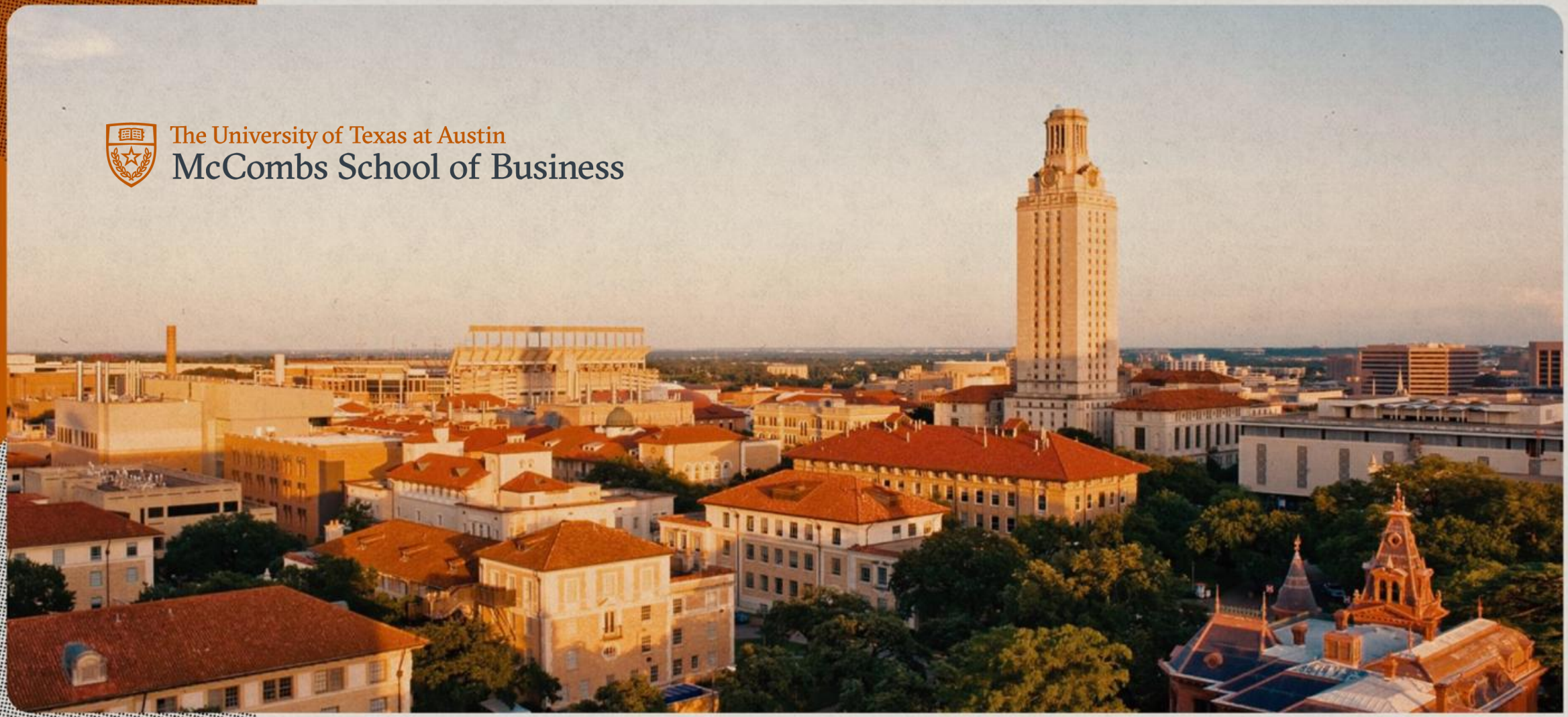


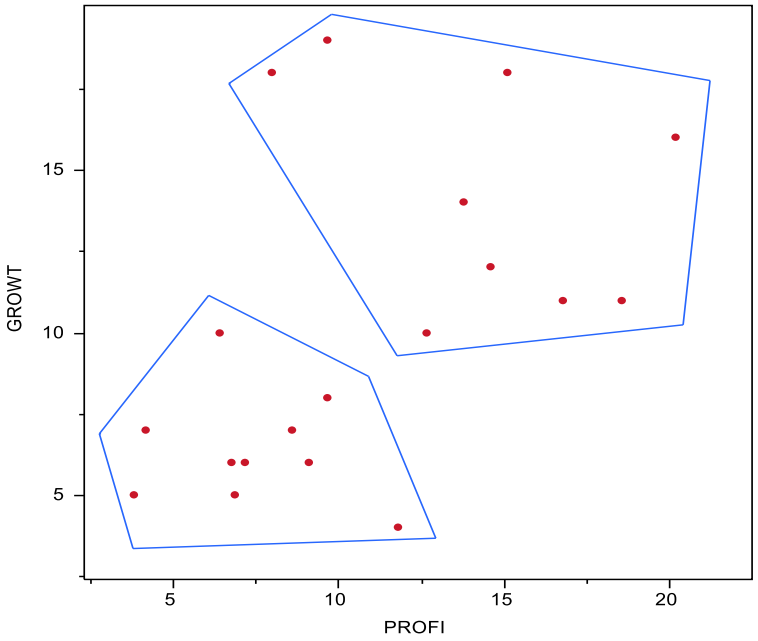
D10: DBSCAN advanced cluster analysis

DBSCAN from 40,000 feet

DBSCAN = **D**ensity-**B**ased **S**patial **C**lustering of **A**pplications with **N**oise

DBSCAN is an advanced CA based on the familiar idea that a cluster is a concentration of points that is separated from other concentrations.

Ex: From CA notes:



DBSCAN from 40,000 feet

DBSCAN offers the new functionality to classify cases as

- core within a cluster
- on the boundary of a cluster
- noise – cases that do not seem to follow the clustering worldview

As with GMM, cases occur

- more frequently where the data probabilities are high: where the probability density function is high;
- less frequently where the data probabilities are low: where the probability density function is low.

Illustration

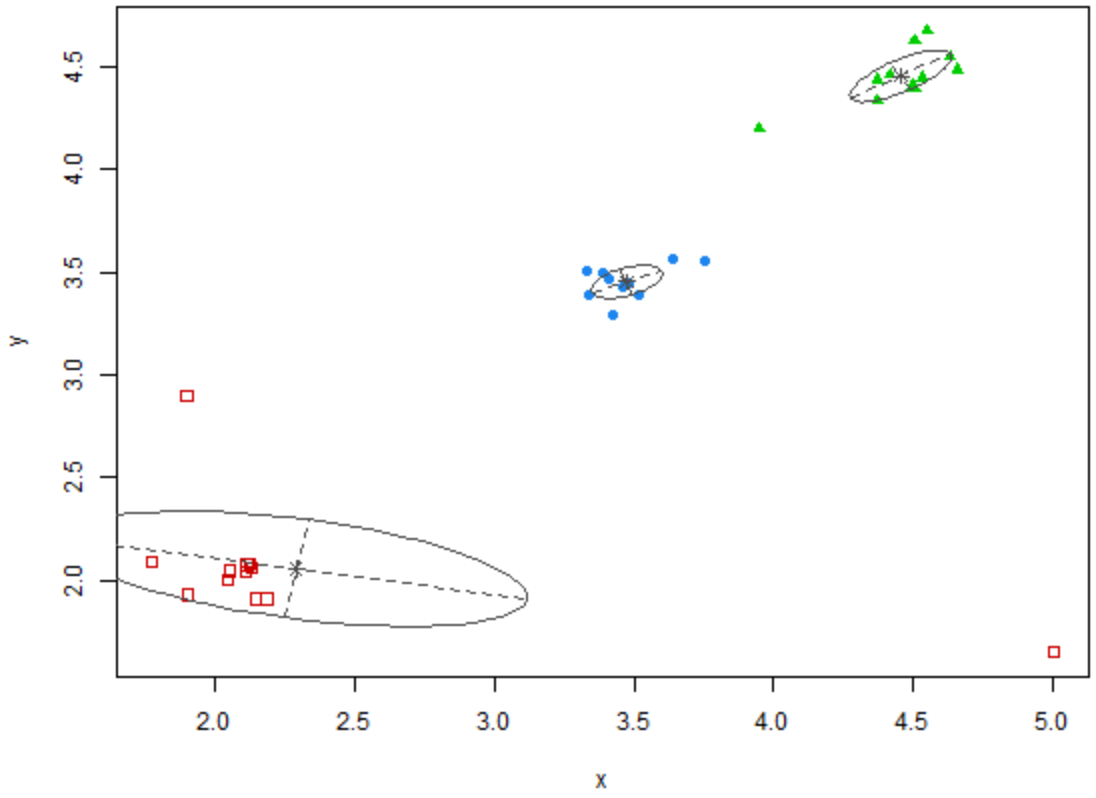
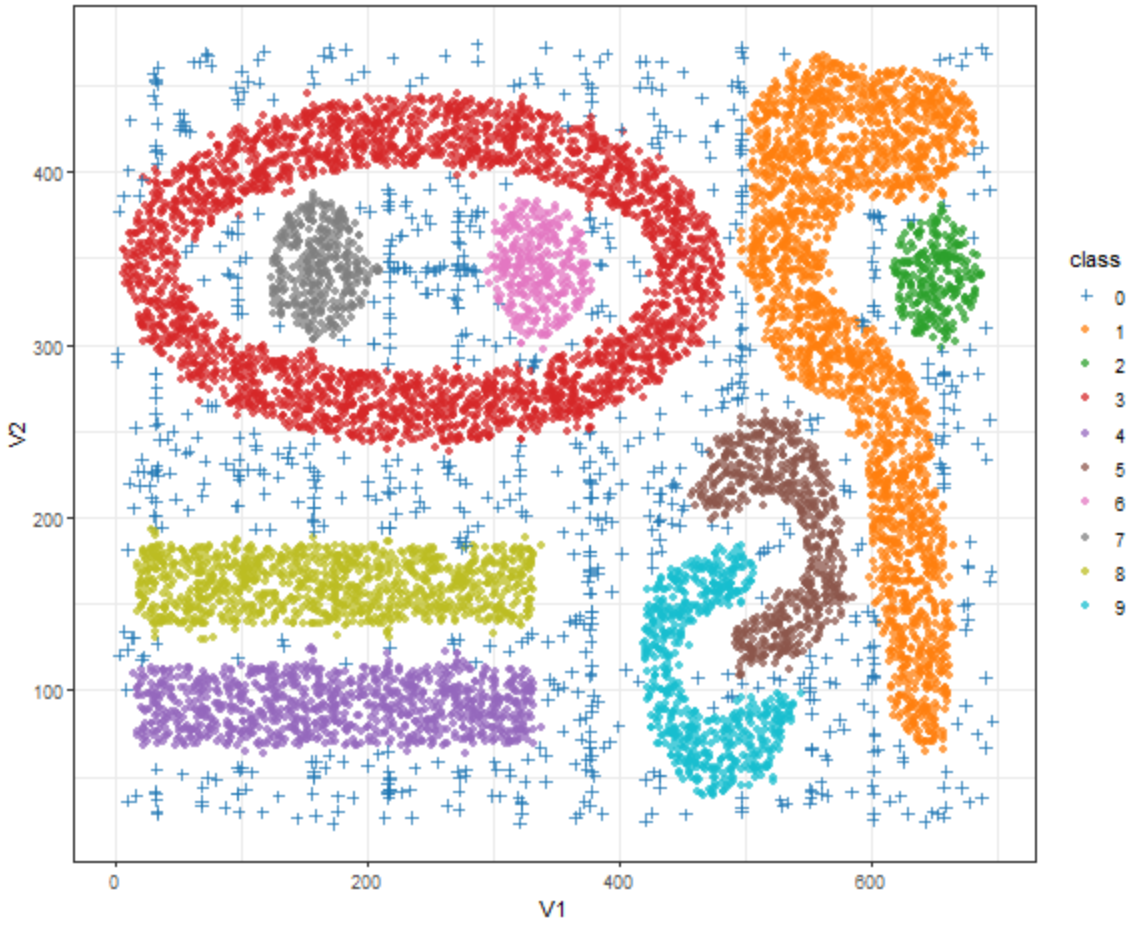
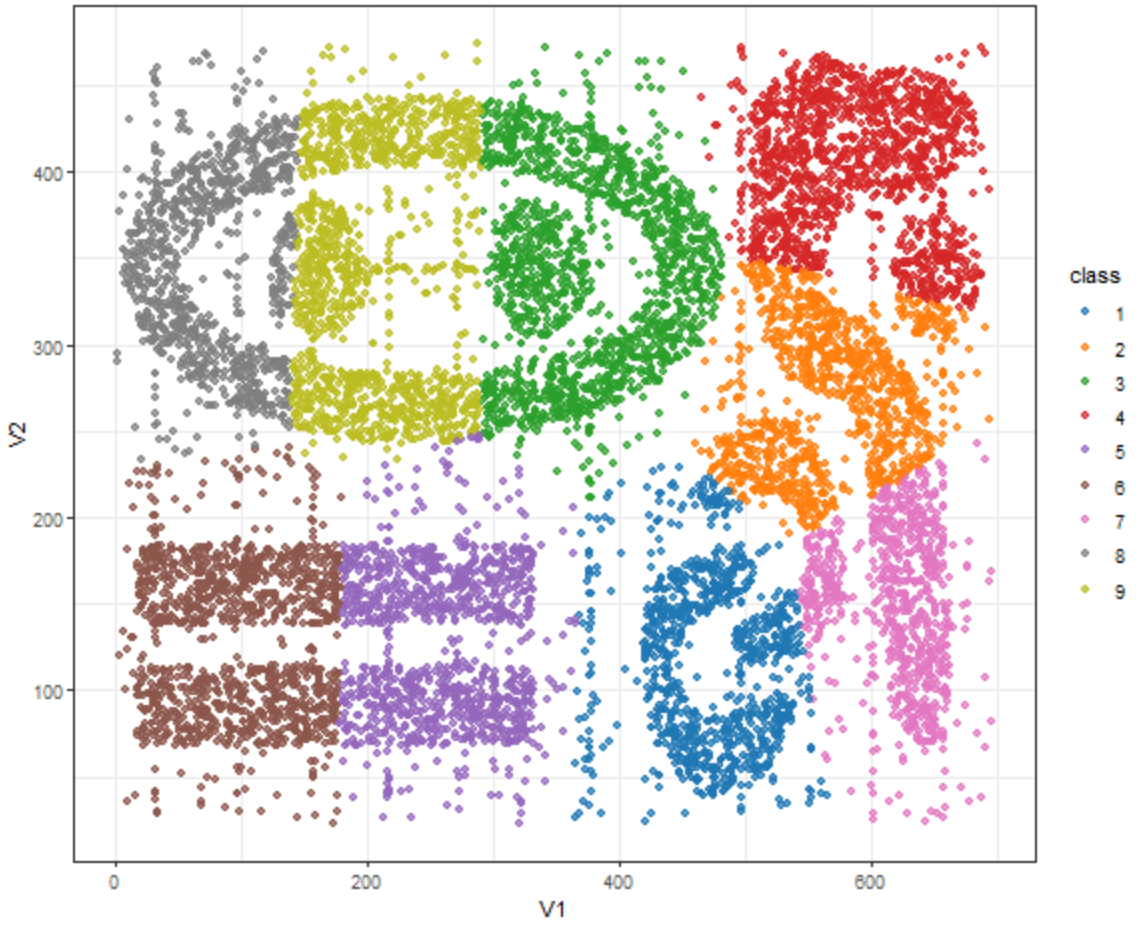


Illustration of DBSCAN result



Compared with K-means on same data



How DBSCAN Works

DBSCAN setup:

```
db <- dbscan(data, eps = eps, minPts = minPts)
```

Two critical parameters: eps, minPts

eps defines the neighborhood in which DBSCAN looks for concentrations of data. eps is the (fixed) radius of those neighborhoods.

minPts is the minimum number of points that it takes to make a concentration.

How DBSCAN Works

Start with a random point in the dataset.

Count all data within eps of that point.

If that count equals or exceeds minPts, call the point "core“.

Start that point as a seed for a cluster.

Then expand the cluster by recursively adding points that are "density reachable" from the seed.

A point is **density reachable** *from the seed if the point is within eps of the seed.*

Then examine the added reachable points. If any reachable point is also core, then add all points that are reachable from it.

How DBSCAN Works

This starts a process of chaining together core points that extend out into the environs of the initial seed. This chaining process will stop when there are no core points within eps of any core points in the nascent cluster.

Then go on to another point that is outside of the first cluster. Test it for being core. If it is core, it becomes a seed for a second cluster. Expand it by adding chains of reachable points. Continue.

How DBSCAN Works

There are three types of points: core, border, and noise

A core point has at least minPts points within eps of itself

A border point is within eps of a core point but is not core itself b/c it has fewer than minPts within eps of itself.

A noise point is neither a core point nor a border point.

Advantages of DBSCAN

Flexibility through choice of parameters eps and minPts

Ability to handle noise

Ability to discover clusters of arbitrary shape

Applications of DBSCAN

Widely used in various fields b/c of ability to discover clusters of arbitrary shape and ability to handle noise

Some examples:

Geospatial data: disease outbreaks, crime outbreaks

Image processing: find pixel concentrations

Marketing: group similar items in retail purchasing data

Limitations of DBSCAN

Sensitive to choice of parameters eps and minPts

Many datasets have clusters of different concentrations and width; DBSCAN uses fixed eps

Curse of dimensionality

Choosing eps and minPts

The choice critically affects the clustering results.

Rules of thumb:

- Set minPts at least 1 + number of analysis variables.
- Set minPts = 2 * number of analysis variables.

Experiment with a range of values for eps and minPts.

(But be wary of overfitting.)

Use any relevant subject-matter knowledge

Use a k-distance graph.

Useful to refine eps once you have a tentative k for minPts.

Choosing eps and minPts

Use a k-distance graph.

Useful to refine eps once you have a tentative k for minPts.

First, calculate the distance from each data point to its kth nearest neighbor, where k is the tentative value for minPts.

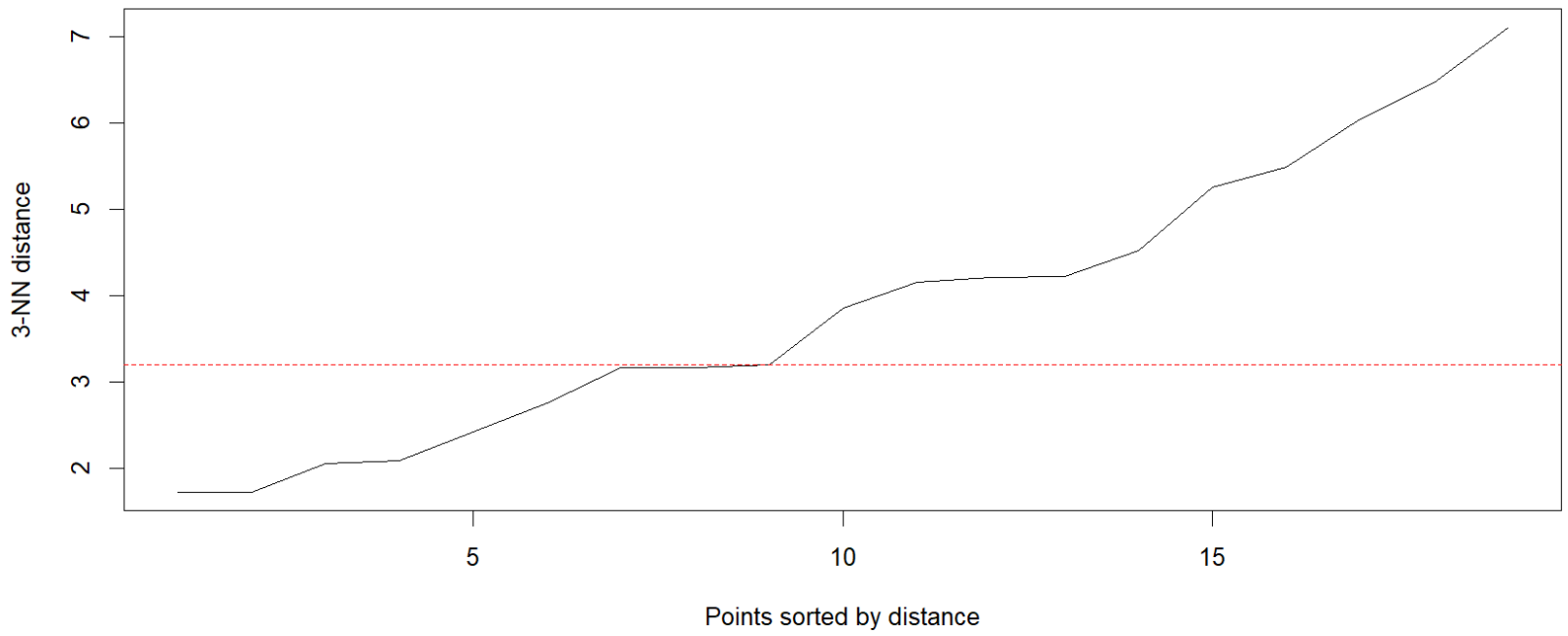
Order these distances from least to greatest.

Plot those distances from 1 to n. The plot will be increasing.

The plot is called a k-distance graph.

Look for an “elbow” in the plot. The elevation (distance) at the elbow is a good choice for eps.

K-distance graph example

For clustering of 19 stocks by two analysis variables, PROFIT and GROWTH.

For k=3 minPts, an eps of about 3.2 is indicated for this elbow.